

Radio Interno

Definición 1 (Radio interno). Sea $\Omega \subset \mathbb{C}$ un dominio, $R(\Omega)$ es el radio interno de Ω

$$\iff R(\Omega) = \sup \{r > 0 : \exists z \in \Omega : D(z, r) \subset \Omega\}.$$

Ejemplos 2 (de radios internos de difentes dominios).

- [1] Para la banda $\Omega_1 = \{z \in \mathbb{C} : -L/2 < \Re z < L/2\}$, se tiene que $R(\Omega_1) = L$.
- [2] Para el semiplano superior $\Omega_2 = \mathbb{H}$, se tiene que $R(\Omega_2) = \infty$.
- [3] Para $\Omega_3 = \mathbb{C} \setminus \{n + mi : n, m \in \mathbb{Z}\}$, se tiene que $R(\Omega_3) = \frac{\sqrt{2}}{2}$.

Observación 3. Sea $\Omega \subset \mathbb{C}$ un dominio, $a \in \mathbb{C}$ y $r > 0$. Entonces, $R(a + r\Omega) = rR(\Omega)$.

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